Shell-Coded Casimir Boundary Transduction: Physical Readout Architectures for the White Sphere–Cylinder Geometry

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Abstract

White et al. introduced a custom Casimir geometry in which a micron-scale sphere inside a conducting cylinder produced a shell-like worldline-numerics vacuum response. This paper develops the physical continuation of that geometry as a boundary-transduction problem. The experimental target is a geometry-locked Casimir response associated with the annular sphere–wall region, measured through ordinary precision apparatus channels with explicit material and electromagnetic controls.

The central object is a shell-coded boundary response: a finite-geometry renormalized-stress organization whose physical evidence appears as force, pressure, resonant frequency, phase, or superconducting microwave response. This framing preserves the useful content of the White sphere–cylinder result while placing the measurable claim inside established Casimir metrology. The metric comparison supplies scale context; the apparatus claim is a measurable boundary response.

Three readout architectures are identified. A paired differential-pressure cell gives the direct stress/force channel and requires strong patch-potential, roughness, membrane, thermal, and vibration control. A high-Q electromagnetic cavity converts boundary response into a frequency, phase, or mode-overlap observable. A superconducting resonator provides a cryogenic microwave path in which shell response can be separated from kinetic inductance, surface loss, vortex loading, quasiparticles, and material controls. Material impedance and capacitance measurements form the calibration channel that maps ordinary apparatus response.

The resulting program is precision finite-geometry Casimir metrology. A positive shell-coherent result would identify a repeatable boundary-conditioned vacuum response in a finite shell-forming geometry and would supply a calibrated transducer for that response. The practical value is in MEMS/NEMS force calibration, patch-potential diagnostics, high-Q boundary sensing, superconducting resonator loss studies, cryogenic gap metrology, and validation data for finite-geometry Casimir calculations.

1. INTRODUCTION

Casimir experiments translate boundary-conditioned quantum fields into measurable apparatus behavior. Conductors, dielectrics, superconductors, and structured finite geometries alter the electromagnetic mode spectrum. The renormalized stress associated with that altered spectrum appears in mechanical force, pressure, force gradients, resonant frequency shifts, phase shifts, quality-factor changes, and temperature-dependent superconducting responses. Contemporary Casimir metrology is therefore an instrumentation discipline as much as a field-theory calculation: it depends on surface potential control, separation metrology, material characterization, roughness models, finite-conductivity corrections, resonator stability, and environmental rejection [4, 9, 5, 7].

White, Vera, Han, Bruccoleri, and MacArthur used worldline numerics to study custom Casimir cavities and reported a sphere-in-cylinder case whose negative energy-density morphology resembles the shell placement drawn in an Alcubierre-style source section [1, 2]. The compact case uses a $1\,\mu$m diameter sphere centered in a nominal $4\,\mu$m diameter cylinder. Its scientific value comes from role structure. The central sphere, inner wall, annular gap, material boundary, exterior control region, and possible readout channel can be assigned separate physical jobs.

This paper treats the White sphere–cylinder system as a finite Casimir boundary object. The question is how the shell-coded boundary response can be read by a real instrument. The answer is a physical architecture with a geometry-indexed response vector. A viable readout must identify a transduction channel, measure material and electromagnetic backgrounds, modulate geometry, fit a shell template against nuisance templates, and give a physical interpretation for the recovered coefficient.

The readout program has three serious apparatus families. Differential pressure reads boundary stress directly through paired force cells. High-Q electromagnetic cavities read boundary response through resonant mode perturbation. Superconducting resonators read cryogenic boundary response through microwave frequency, phase, loss, and nonlinear mechanical channels. These branches are supported by existing experimental machinery: MEMS/NEMS Casimir force sensors [4, 7], high-Q nanomembrane force probes with Kelvin-probe surface-potential control [5, 6], long-range differential force measurements with explicit patch and roughness accounting [9], and superconducting Casimir or microwave-resonator platforms [8, 10, 11].

2. THE PHYSICAL CLAIM

The shell-coded response is a statement about renormalized stress-energy in a finite boundary value problem. In QFT, local energy density is represented by the renormalized expectation value of the stress-energy tensor. Boundary conditions can make that local value lower than the reference vacuum in some regions while averaged constraints keep the physical system bounded. Ford's review describes this operational status of negative energy densities and the quantum inequality restrictions that accompany them [3].

Casimir apparatus makes this concept measurable. The negative sign in the ideal parallel-plate energy density expresses a boundary-conditioned stress relative to the reference state. The physical content is observed through force, pressure, energy, and mode behavior. Finite geometries add spatial structure: a boundary can concentrate the response in a specific region and the apparatus can couple to that region with varying strength.

The White sphere–cylinder geometry contributes a candidate spatial code. Its annular sphere–wall region acts as the shell role. The physical readout program asks whether that role can be coupled to a pressure cell, a cavity mode, or a superconducting resonator with enough strength and template separation to survive ordinary material and electromagnetic controls.

3. RELATION TO WHITE ET AL.

White et al. supplied a geometry, a worldline calculation, and a provocative metric comparison [1]. The present analysis separates the pieces needed for experimental use:

1. Geometry: a small, finite, role-structured sphere–cylinder boundary.
2. Morphology: a shell-like organization of the vacuum-boundary response.
3. Scale: the calibrated stress level associated with that response.
4. Transduction: the physical apparatus channel that converts the response into measured data.

The first two pieces motivate laboratory design. The third piece supplies metric-scale context. The fourth piece defines the experimental claim. In this form, the White geometry becomes a test object for finite-geometry Casimir metrology: if the shell response is physical at an apparatus scale, it should appear as a geometry-locked coefficient in a force, pressure, cavity, or superconducting resonator dataset.

4. READOUT MODEL

Let q denote a controlled geometry or apparatus setting: gap, sphere radius or surrogate radius, cylinder radius, wall thickness, lateral offset, cell length, mode overlap, readout radius, temperature, magnetic history, or drive power. The measured observable $y(q)$ is modeled as

$$y(q) = a_{\text{shell}} T_{\text{shell}}(q) + \sum_i a_i T_i(q) + \varepsilon(q), \tag{1}$$

where T_{shell} is the shell-coded response template and the T_i are measured background templates. The background family includes patch potential, capacitance, roughness, membrane drift, temperature, vibration, finite conductivity, waveguide dispersion, mode mixing, kinetic inductance, surface loss, vortex response, quasiparticles, and material controls.

Equation 1 defines the evidential standard. The result is a shell coefficient with uncertainty, a set of nuisance coefficients with uncertainties, and a false-positive check from material-only and electromagnetic-only control schedules. A single absolute shift has limited meaning in this apparatus class. A geometry-locked response vector can separate shell, material, and readout behavior.

Table 1: Physical readout families for the shell-coded boundary response.

Family	Observable	Physical transduction	Main control role
Differential pressure	Paired pressure or force contrast	Boundary stress in the shell-gap geometry	Patch, roughness, membrane, thermal, and vibration rejection
High-Q cavity	$\delta f/f$, phase, Q, or mode splitting	Electromagnetic eigenmode perturbation from shell-sensitive boundary overlap	Loss, thermal, waveguide, and mode-mixing separation
Superconducting resonator	Microwave $\delta f/f$, phase, Q, nonlinear drum response, or T_c-coupled response	Cryogenic boundary response in a superconducting microwave apparatus	Surface loss, kinetic inductance, vortex, quasiparticle, and material separation
Material impedance	Capacitance, impedance, contact, or loss response	Ordinary material and electromagnetic apparatus behavior	Calibration map for backgrounds

The design-screening ladder associated with this paper used a recovery gate of SBR = 1.2 and an electromagnetic-only false-positive target near FP < 0.05. Table 2 summarizes the portable design implications.

5. COMMON APPARATUS ARCHITECTURE

All three readout branches require the same experimental skeleton. The apparatus contains a shell-sensitive cell, a matched dummy cell, a material-control channel, a geometry actuator, surface metrology, and a readout chain with environmental monitors. The shell cell implements the sphere–cylinder

Table 2: Screening-level design constraints for the physical branches.

Branch	Representative viable cases	Predicted SBR range	Design implication
High-Q cavity	nominal clean through optimistic clean/matched/lossy	2.22–19.34	Strongest design row; mode overlap and loss templates set interpretation
Superconducting resonator	nominal balanced/clean through optimistic clean/balanced/lossy	3.59–17.76	Strong cryogenic path; surface-loss and quasiparticle controls are central
Differential pressure	optimistic balanced and optimistic quiet	2.02–5.31	Direct force path; patch and common-mode controls set feasibility
Material impedance	calibration rows	background map	Measures ordinary apparatus response for the same geometry

geometry or a manufacturable surrogate preserving the same shell-overlap coordinates. The dummy cell preserves material, wiring, thermal mass, and readout coupling while suppressing shell overlap. The material channel measures impedance, capacitance, contact, and loss behavior across the same schedule.

The geometry schedule is part of the measurement. Gap, radius, offset, wall thickness, length, and readout overlap should move the shell template differently from patch, capacitance, material, and drift templates. Surface measurements should accompany those settings. Garcia-Sanchez et al. show how Kelvin-probe surface-potential mapping can be integrated into high-Q nanomembrane Casimir measurements [5]. Bimonte et al. demonstrate a long-separation force measurement with explicit treatment of patch potentials, roughness, edge effects, uncertainty, and theory comparison [9].

The instrument must first demonstrate ordinary Casimir sensitivity in a known geometry. This calibration step establishes separation metrology, electrostatic calibration, roughness priors, and noise floors. The shell geometry then becomes a finite-boundary test object whose output is judged by template selection and control behavior.

6. DIFFERENTIAL-PRESSURE ARCHITECTURE

The differential-pressure architecture reads boundary stress directly. Two matched cells share environment and readout while their geometry settings produce different shell-gap responses. The primary observable is

$$\Delta P(q) = P_{\mathrm{shell}}(q) - P_{\mathrm{control}}(q), \tag{2}$$

with the fit model of Eq. 1. For ideal parallel plates,

$$P_{\mathrm{C}}(a) = -\frac{\pi^2 \hbar c}{240 a^4}, \tag{3}$$

which gives about 1.3×10^{-3} Pa at $a = 1\,\mu$m. The sphere–cylinder cell is finite and curved, so Eq. 3 serves as a scale reference for the apparatus model. The screening rows place the strongest pressure cases in the same metrological neighborhood, with shell pressures around 10^{-4}–10^{-3} Pa.

The machinery is close to established Casimir force metrology. Decca, Aksyuk, and Lopez review micro- and nano-electromechanical systems for Casimir measurements [4]. Stange et al. show a practical commercial MEMS force platform with piconewton-scale force resolution and electrostatic calibration [7]. Garcia-Sanchez et al. demonstrate high-Q nanomembrane force-gradient readout with in situ surface-potential control [5, 6]. These examples define the required toolchain: calibrated actuation, separation sensing, patch mapping, roughness measurement, and common-mode rejection.

A pressure result would identify the shell-coded boundary response as a finite-geometry force or stress contrast. The immediate utility is precision microforce calibration, patch-potential diagnostics, roughness validation, and MEMS/NEMS pressure metrology at micron gaps.

7. HIGH-Q CAVITY ARCHITECTURE

The high-Q cavity architecture uses resonance as the transducer. A cavity mode with large field overlap in the shell-sensitive region converts a small boundary perturbation into a measurable frequency, phase, Q, or mode-splitting signal. In first-order perturbation language,

$$\frac{\delta f}{f} \sim -\frac{1}{2}\frac{\int_\Omega \left(\delta\epsilon\,|E|^2 - \delta\mu\,|H|^2\right)\,dV}{\int_\Omega \left(\epsilon|E|^2 + \mu|H|^2\right)\,dV} + \Delta_{\text{boundary}}, \tag{4}$$

where Δ_{boundary} denotes the shell-sensitive finite boundary contribution modeled by the apparatus eigenmode solver. The equation states the design requirement: maximize known shell overlap and measure the ordinary material terms in the same basis.

The screening ladder ranks this branch highly. The strongest cavity row gives SBR = 19.34, and the nominal clean row remains above the recovery gate. Those values require measured templates for conductor loss, thermal expansion, waveguide dispersion, port coupling, and mode mixing. A strong cavity design therefore includes a shell-overlap mode, a low-overlap or null mode, a dummy cavity, independent thermal metrology, deliberate port-coupling sweeps, and a validated eigenmode model.

The positive interpretation is electromagnetic mode transduction of the shell-coded boundary response. Such a result would support high-Q boundary-sensitive sensors, finite-geometry cavity calibration, and experimental validation of Casimir calculations in resonant structures.

8. SUPERCONDUCTING-RESONATOR ARCHITECTURE

The superconducting architecture uses a cryogenic microwave resonator, drum, or transition-sensitive cavity to read the shell-coded response. The observable can be frequency shift, phase shift, Q change, nonlinear mechanical response, or a superconducting transition response. The core fit is

$$\left(\frac{\delta f}{f}\right)_{\text{sc}} = a_{\text{shell}}T_{\text{shell}} + a_{\text{ki}}T_{\text{ki}} + a_{\text{surf}}T_{\text{surf}} + a_{\text{vort}}T_{\text{vort}} + a_{\text{qp}}T_{\text{qp}} + a_{\text{mat}}T_{\text{mat}} + \varepsilon, \tag{5}$$

with kinetic inductance, surface loss, vortex loading, quasiparticles, and material response treated as measured templates.

This branch has unusually strong source support. Perez et al. designed a NEMS system to probe Casimir energy corrections to superconducting condensation energy by measuring a superconducting film in a controlled cavity [8]. de Jong et al. report a superconducting Casimir force measurement through nonlinear dynamics of a superconducting drum resonator in a microwave optomechanical system [11]. Gurevich reviews the microwave-loss mechanisms that govern superconducting resonators, including quasiparticles, residual surface resistance, dielectric loss, trapped vortices, kinetic inductance effects, and high-field limits [10]. Those mechanisms are the exact nuisance family in Eq. 5.

The screening ladder gives superconducting rows from SBR = 3.59 to SBR = 17.76 across the viable nominal and optimistic cases. The apparatus should include shell-coupled and material-only resonators on the same cooldown path. Temperature sweeps across and below T_c, magnetic-history tests, vortex-loading controls, drive-power sweeps, quasiparticle checks, and geometry modulation all become evidence-bearing coordinates.

A superconducting result would identify a cryogenic boundary response separated from microwave-loss physics. Its practical value is superconducting surface-loss diagnostics, cryogenic vacuum-gap metrology, microwave boundary-sensing, and experimental tests of superconducting electrodynamics in finite Casimir geometries.

9. CONTROL MATRIX AND OUTCOME CLASSES

The control channels are measured observables. They define the interpretation of the shell coefficient. Table 3 summarizes the control standard for each physical branch.

Table 3: Evidence-bearing controls for physical readout branches.

Branch	Required controls	Shell-coherent data signature
Differential pressure	Matched dummy cell, opposite gap schedule, Kelvin-probe map, roughness prior, membrane drift model, thermal and vibration monitors	Pressure vector selects the shell schedule after patch, roughness, and membrane templates are included
High-Q cavity	Eigenmode model, shell-overlap and low-overlap modes, dummy cavity, thermal expansion monitor, loss and port-coupling sweeps	Frequency or phase vector selects shell overlap over loss, thermal, waveguide, and mode-mixing templates
Superconducting resonator	Material-only resonator, temperature sweep, drive-power sweep, magnetic-history control, quasiparticle checks, surface-loss model	Microwave response selects shell schedule after kinetic-inductance, loss, vortex, quasiparticle, and material templates are included

Table 4: Outcome classes for shell-geometry measurements.

Outcome	Data signature	Physical meaning
Shell coherent	Shell coefficient exceeds recovery gate and controls satisfy the false-positive target	Finite White-style boundary geometry produces a measurable shell-coded Casimir response in that transducer
Material selected	Patch, capacitance, impedance, contact, or loss template has the stable coefficient	The apparatus maps ordinary material/electromagnetic response of the same geometry
Resonator selected	Cavity or superconducting response follows mode mixing, conductor loss, surface loss, vortex, or quasiparticle template	Resonator physics dominates the readout channel under the tested settings
Sensitivity bounded	Shell and nuisance templates remain below recovery gate after calibration	The apparatus sets an upper bound on shell-coded transduction for that branch

Every outcome is useful. Shell coherence opens a dedicated apparatus design paper. Material or resonator selection improves the background map. A sensitivity bound sets a quantitative design target for the next transducer.

10. PHYSICAL IMPLICATIONS

A pressure result would identify a direct finite-geometry stress response. It would connect the shell-coded morphology to a mechanical observable and would test how patch potentials, roughness, edge structure, and common-mode rejection interact with finite Casimir geometry.

A high-Q cavity result would identify resonant electromagnetic transduction. The shell response would appear as an eigenmode perturbation with measurable geometry dependence. This path gives the strongest design-screening row and offers a natural route to compact boundary sensors.

A superconducting resonator result would identify a cryogenic boundary response separated from superconducting loss channels. This branch is especially valuable because superconducting Casimir force measurements, condensation-energy questions, microwave resonator loss, and quantum-limited sensing share one experimental language.

Across all branches, the physics is finite-boundary QFT metrology. The measured quantity is the consequence of boundary-conditioned renormalized stress as read through a calibrated apparatus channel. The metric comparison remains a scale reference and a motivation for the shell geometry.

11. PRACTICAL UTILITY

The program has practical utility independent of its origin in the White geometry. A working pressure platform supports micron-gap force standards, MEMS/NEMS calibration, and surface-patch diagnostics. A working cavity platform supports high-Q sensors for boundary motion, material changes, and finite-geometry electromagnetic perturbations. A working superconducting platform supports cryogenic gap metrology, microwave-loss diagnostics, vortex and quasiparticle characterization, and superconducting Casimir tests.

The schedule-and-template method is broadly reusable. Any finite Casimir geometry with correlated material and readout backgrounds can be treated as a template-selection problem. The result is a clearer separation between geometry, material response, and readout physics.

12. REPRODUCIBILITY AND SOURCE BASIS

The citable scientific basis for this manuscript is the public literature in the reference list. The references provide the canonical DOI, arXiv, NIST, and journal locations for the source material.

The numerical screening values quoted in Tables 2 and 3 come from a finite readout-ladder calculation using shell-template recovery, electromagnetic-only false-positive checks, and physical backend cases for pressure, cavity, superconducting, material, and central-reference channels. The associated supplementary material consists of compact machine-readable screening summaries: candidate definitions, backend physical cases, gate summaries, and run-level summary metadata. Archiving the manuscript source and PDF with those summaries keeps the quoted design-screening values traceable.

13. CONCLUSION

The White sphere–cylinder geometry is a finite Casimir boundary object with a shell-coded response morphology. Its productive experimental continuation is physical transduction: reading the shell response through force, pressure, high-Q electromagnetic modes, or superconducting microwave modes while material and readout backgrounds are measured in the same apparatus.

Three architectures define the program. Differential pressure gives the direct stress channel. High-Q cavities give resonant electromagnetic amplification. Superconducting resonators give a cryogenic microwave path with rich loss-separated controls. Together they convert a shell-forming Casimir geometry into a buildable metrology problem with clear outcomes, portable controls, and practical value for finite-geometry vacuum-boundary physics.

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